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Validation Report
AI - End-to-End Pipeline Validation
v1.0

**End-to-End Validation of the
Glaucoma Detection Pipeline**

Project Subject: Glaucoma detection using intelligent
analysis of fundus images

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1 Introduction

The ROI detection (YOLO) and optic disc/cup segmentation (U-Net++) stages have each been documented separately (see the related team reports in Section 6). This report addresses the remaining question: how well does the *complete* pipeline perform as a glaucoma *classifier*, end to end, rather than as a set of individually strong components.

The pipeline is evaluated as deployed: a full, uncropped fundus photograph is passed through YOLOv9 ROI detection and cropping, U-Net++ (EfficientNet-B4) disc and cup segmentation, area cup-to-disc ratio (aCDR) computation, and a single-threshold decision. The diagnostic metrics reported here—confusion matrix, ROC/AUROC, sensitivity and specificity—were not covered by the earlier component reports.

2 Validation Setup

Dataset. The segmentation model was trained on the multi-source SMDG benchmark: a class-balanced collection of approximately 1,300 fundus images carrying expert disc and cup masks ($\approx 50/50$ glaucoma/healthy), split 80/20 into training and validation. This supersedes the earlier single-dataset baseline (1020 images, no augmentation; documented in the *U-Net++ Baseline* report), which the model has since evolved beyond.

Held-out validation set. All metrics below are computed on the 267-image held-out validation split, never seen during segmentation training. Its composition by source database and class is given in Table 1.

End-to-end protocol. Whereas the segmentation network was trained and monitored per epoch on ROIs already cropped by YOLOv9, here the same validation images were submitted as *full, uncropped* photographs and processed by the entire pipeline. The evaluation therefore includes the ROI-detection stage and reflects real deployment rather than an oracle crop.

Table 1: Composition of the 267-image held-out validation set.

Source	Healthy	Glaucoma	Total
G1020	29	51	80
ORIGA	30	38	68
REFUGE1	53	12	65
PAPILA	19	18	37
DRISHTI-GS	1	10	11
CRFO	0	6	6
Total	132	135	267

3 Segmentation Quality on the Validation Set

On the held-out validation crops the model reaches a mean Dice of 0.922 (disc Dice 0.953, IoU 0.910, pixel accuracy 97.9%; cup Dice 0.892, IoU 0.804, pixel accuracy 98.6%). Figure 1 shows the mean Dice per source database.

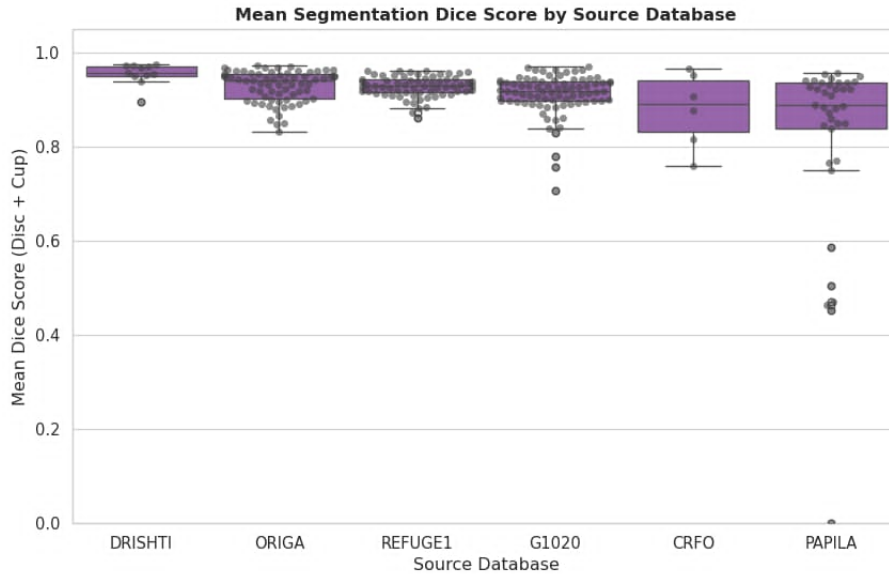


Figure 1: Mean segmentation Dice (disc + cup) per source database within the validation set.

Conclusion. Segmentation quality is high and consistent across all six databases (mean Dice ≈ 0.89 - 0.96), indicating that the model generalises across imaging conditions rather than overfitting one collection. The disc is segmented more accurately than the cup, whose boundary has lower contrast. On the full, uncropped end-to-end run the cup Dice falls slightly (to ≈ 0.86 , mean ≈ 0.91) because the ROI is produced by YOLOv9 rather than taken from ground truth, while disc Dice is unchanged. Segmentation is robust and is *not* the limiting factor of the system.

4 End-to-End Classification

The decision threshold was calibrated on the ROC curve to the balanced-performance operating point ($aCDR > 0.323$), at which the end-to-end AUROC is 0.704 (Figure 2). The confusion matrix at this threshold is given in Table 2.

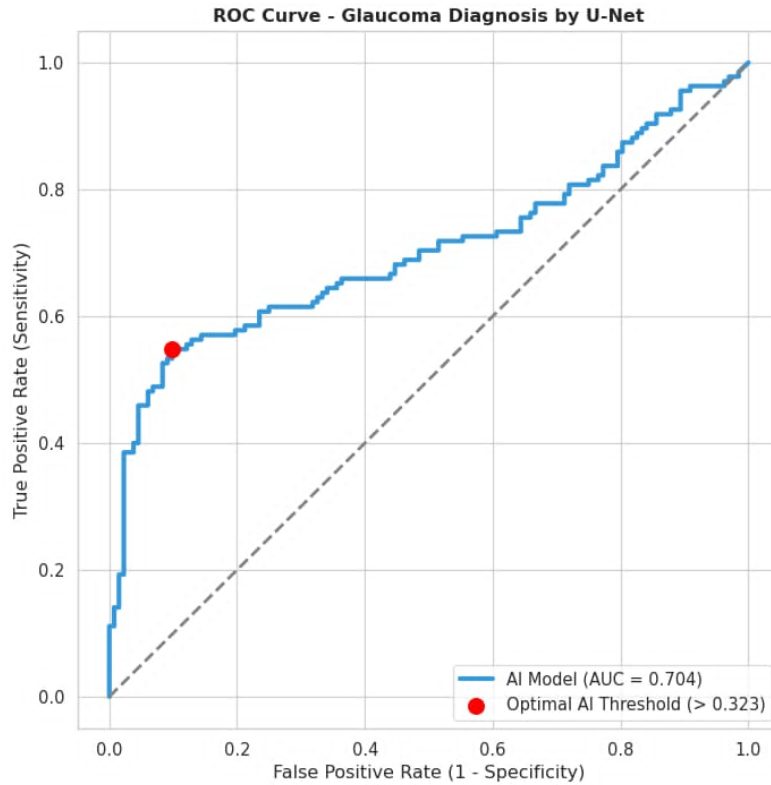


Figure 2: ROC curve for end-to-end glaucoma classification (AUROC = 0.704). The marked point is the operating threshold $\text{aCDR} > 0.323$.

Table 2: End-to-end confusion matrix ($n = 267$, threshold $\text{aCDR} > 0.323$).

	Pred. Healthy	Pred. Glaucoma
Actual Healthy	119 (TN)	13 (FP)
Actual Glaucoma	61 (FN)	74 (TP)

The resulting operating-point metrics are:

- Accuracy: 72.3%
- Sensitivity (recall): 54.8%
- Specificity: 90.2%
- Precision: 85.1%
- F1-score: 66.7%
- AUROC: 0.704

Conclusion. The system is highly specific (90.2%) and precise (85.1%) but insufficiently sensitive (54.8%): it rarely raises false alarms, yet misses a large share of true glaucoma cases (61 false negatives out of 135). For a screening tool, sensitivity is the decisive metric, so this profile-high specificity, low sensitivity-is the central finding and defines the problem to be solved. The companion analysis report (Section 6) isolates its cause

and shows that it stems from the single-threshold decision rule rather than from the segmentation network.

5 Summary

- The full pipeline was validated end to end on 267 held-out, multi-source images fed as full, uncropped photographs.
- Segmentation is strong and consistent across databases (mean Dice 0.922) and is not the bottleneck.
- End-to-end classification is specific (90.2%) but not sensitive (54.8%); accuracy 72.3%, AUROC 0.704.
- The high-specificity / low-sensitivity profile defines the problem; its root cause is analysed in the companion report.

6 Related Team Documents

- *Comparative Analysis of YOLO Models for ROI Detection* - the YOLOv9 selection used as the ROI stage here.
- *U-Net++ Training Results (Baseline, No Augmentation)* - the earlier 1020-image baseline that the current multi-source model evolved from.
- *CDR Biomarker Reliability and the Single-Threshold Bottleneck* - companion analysis that localises the cause of the limited sensitivity.